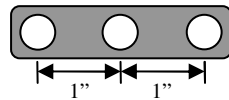
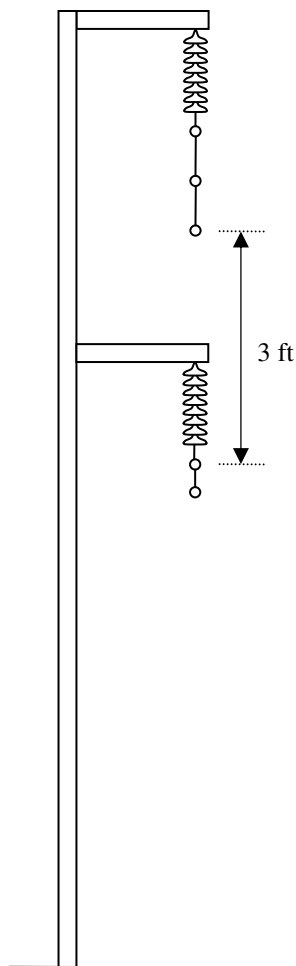


An insulated conductor three phase power cable uses three identical solid copper wires of 0.125" radius encased in a polyvinyl insulating material as shown below. Calculate the **distributed per phase series inductance** of this power cable **in units H/kft**. (Hint: 1 mile = 5280 ft = 5.28 kft)



$$L = 156.088 \mu\text{H/kft}$$

A open conductor overhead single phase transmission line uses 3 ACSR “*Leghorn*” cables with 8” spacing in one conductor and 2 ACSR “*Ostrich*” cables with 5” spacing in the other conductor as shown in the support tower diagram below. The cable data is also given below. Calculate the **distributed series resistance** of the transmission line **at a temperature of 30°C**, and also calculate the **geometric mean distance** between the transmission line conductors.



$$R = 0.30947 \Omega/\text{mi at } 30^\circ\text{C}$$

$$\text{GMD}_{xy} = 3.8305 \text{ ft}$$